Combinatorics (Additive, extremal, graph, yayayayayay)

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Yeah so a lot of combinatorics is about finding valid constructions. Actually this is a trend in all of math, so a background in what a construction is is great.

1 Set Theory

2 Extremal Combinatorics

Definition

Sunflower A sunflower is a collection of sets $S_1, S_2, \ldots, S_k$ such that $S_i \cap S_j$ is constant over all i, j.

It's called a sunflower because it looks like one when you draw it:

Theorem

Erdos-Rado Sunflower Lemma Given a family $\mathcal{F}$ of size s sets, we may find a sunflower if $|\mathcal{F}| > k!(s-1)^k$

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3 Standard Olympiad Combinatorics

3.1 Invariant

3.2 Bounding

4 Graphs

5 Additive Combinatorics